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A. Purpose: To provide guideline instruction for rational use of blood.

B. Scope: Hospital wide

C. Policy:

1. Request

a. To ensure patients safety, blood / blood components should not be prescribed unless there is a real indication.
b. Request should be made by a consultant/doctor working in the hospital.
c. Blood transfusion request form should be filled completely by a doctor
d. Consent for transfusion should be taken from patient / guardian after explaining the transfusion requirement or doctor can give consent in case of unaccompanied patients who are incapable of giving consent.
e. Blood sample should be taken for ABO & Rh grouping and cross matching and labeled at bedside.
f. The entire request for blood should be sent to the Blood Bank.

2. Grouping

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a. All patients should be grouped in case:
i. Any intermediate or major surgery is planned
ii. Any invasive procedures are planned where a risk exists such that blood transfusion is a possibility.
iii. Transfusion of blood is planned
iv. The patient is diagnosed with a medical disease with likelihood of blood transfusion requirement.

3. Blood Reservation

- Blood should be reserved before all elective surgeries. The procedure will be as under:
- The requisition for blood along with the consent form duly signed by the recipient and the blood sample will be sent to the blood storage center immediately on admission / when the need for transfusion is established.

- Cross matching and issue of blood will be done.
- In case of rare blood groups ('AB' Rh positive and all Rh negative blood groups), the
- Treating consultant will discuss with the In Charge of blood bank before the patients admission in order that suitable arrangements can be made in advance.

4. Issuing Blood

- Blood is issued from the blood bank only after compatibility testing which for red cells normally would minimally take 60 minutes. However for all planned surgeries and requirements the requests for compatibility should be sent well in advance.

- Blood compatibility slip will be issued by the Blood bank along with all blood products. This has to be kept in patients file and responsibility of the document will be that of the in charge nurse of the floor / ward / OT.

- After completion of the transfusion, transfusion feedback is filled

- The date of collection and expiring of all unit of blood / FFP / Cryo etc. will be mentioned on all units & these will be issued as per inventory maintenance, normally 'first in-first out' basis in order to optimize blood usage.

5. Procedure Before Transfusion

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a. Blood should be checked by the doctor and the following details should be verified: blood bag with patients blood group, Name, Patient's hospital registration number for correct identification of recipient.

b. Check BP / Pulse / Temperature and record in the case file before transfusion.

6. PROCEDURE DURING TRANSFUSION

a. Monitoring of vitals can be done by nurses

b. Visual observation is often the best way of accessing the patient during transfusion.

c. Record baseline observations at the start of each unit and of each transfusion. Temperature/ Pulse should be measured 15 minutes after the start of each unit and Hourly thereafter.

Monitor rate of flow to ensure transfusion progress. Under no circumstances should any drug be administered through the same IV line.

d. Management of Blood transfusion reactions:

i. Step 1 - Stop transfusion
ii. Step 2 - Keep IV line open with 0.9 % NaCl
iii. Step 3 - Notify attending physician and blood bank
iv. If transfusion is terminated

i- Send freshly collected post – transfusion sample of blood (preferably from opposite arm) and sample of urine to blood bank.
ii- Send the residual blood component unit along with administration set to blood bank.
iii- Fill in the adverse blood transfusion form and forward the same to the Head of clinical audit committee presently, medical superintendent. The committee evaluates the same to determine the cause of reaction, recommend appropriate actions for prevention of the same in future.

Reaction	Signs and symptoms	Aetiology	Clinical action
Allergic	Pruritus Urticara (Hives)	Antibodies to platelets, plasma, protein.	▪ Follow step 1-3 ▪ Administer antihistamines.(oral IM or IV)

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			▪ Resume transfusion if improved ▪ If no improvement after 30 mins treat as under
Allergic or Febrile	Anxiety,pruritus,fever,chills,agitations, flusing,hives,tachycardia,milddyspnoea, hypotension,anaphylaxis.	Antibodies to platelets, plasma, proteins usually IgA	▪ Follow step 1-5 ▪ Administer antihistamine, antipyrectics, vasopressors and corticosteroids as needed.
Acute Hemolytic	anaphylaxis ,anxiety, chestpain, headache, dyspnoea, chills,fever,red urine,agitation,hypotension	Gram negative sepsis, hemolytic	▪ Follow step 1-5 ▪ Treat shock with vasopressors,IV fluids and corticosteroids as needed.
	unexplained bleeding , hemoglobinaemia	transfusion reaction	▪ Maintain adequate airway ▪ Increase renal blood flow with IV fluid ,furosemide , mannitol,maintain brisk diuresis. ▪ If DIC is present consider heparinisation. ▪ Administer Blood components (FFP,Platelets etc) as needed.

e. If no reactions complete transfusion, fill the feedback form and send it to the blood storage center.

f. Whole blood / packed cells can be transfused over 3 – 4 hours.

g. FFP / Cryo: Can be issued within 30 minutes, Should be transfused immediately, If not used can't store again, will not be accepted by the blood storage center.

h. Blood should only be warmed using a specifically designed commercial device with a visible thermometer and audible warning. Blood must not be warmed by insertion in hot water, microwave or on a radiator.

i. Blood giving set must be changed every 2 units or at least every 12 hours.

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